

Unit 1

Solutions

MULTIPLE CHOICE QUESTIONS (MCQS) (ONE MARK)

1.1 Types of Solutions;

1. Camphor in nitrogen gas, is an example of _____. (E)
 - a) liquid Solutions
 - b) solid Solutions
 - c) gaseous Solutions
 - d) aqueous solution
2. Out of following which one is not an example of a solution. (E)
 - a) Air
 - b) Brass
 - c) amalgam
 - d) benzene in water
3. Match the following Type solution with their examples: (E)

Type of solution	Example
A. Gaseous solution	p. dissolved oxygen in drinking water
B. Liquid solution	q. amalgam of mercury with sodium
C. Solid solution	r. atmospheric air
	s. Silicon

- a) A – q; B – r; C – p
 - b) A – r; B – p; C – q
 - c) A – q; B – s; C – p
 - d) A – s; B – r; C – q
4. Match the items given in Column I with the type of solutions given In Column II. (A)

	Column I		Column II
(i)	Soda water	A.	A solution of liquid in solid
(ii)	Sugar solution	B.	A solution of gas in gas
(iii)	German silver	C.	A solution of solid in liquid
(iv)	Air	D.	A solution of solid in solid
		E.	A solution of gas in liquid

- a) (i) - B; (ii) - C; (iii) – D; (iv) – E;
 - b) (i) - E; (ii) - C; (iii) – D; (iv) – B;
 - c) (i) - B; (ii) - E; (iii) – D; (iv) – B;
 - d) (i) - E; (ii) - C; (iii) – B; (iv) – D;

1.2 Expressing concentration of solutions;

5. The concentration term depends on temperature is (E)
 - a) ppm
 - b) mole fraction
 - c) molality
 - d) molarity
6. Which of the following is dependent on temperature? (NEET-17) (E)
 - a) Molarity
 - b) Mole fraction
 - c) Weight percentage
 - d) Molality

7. Increasing the temperature of an aqueous solution will cause (A)
 - a) decrease in molality
 - b) decrease in molarity
 - c) decrease in mole fraction
 - d) decrease in mass percentage (w/w)
8. The mass percentage (w/w) of glucose in water is 10% means (A)
 - a) 10g of glucose dissolved in 100g of water
 - b) 10g of glucose dissolved in 10g of water
 - c) 10g of glucose dissolved in 90g of water
 - d) 10g of glucose dissolved in 90ml of water
9. The mass percentage(w/v) of glucose in water is 10% means. (E)
 - a) 10g of glucose dissolved in 100 g of water
 - b) 10g of glucose dissolved in 90 mL of water
 - c) 10g of glucose dissolved in 100 mL of water
 - d) 10g of glucose dissolved in 90 g of water
10. The mole fraction of the solute in one molal aqueous solution is (D)
 - a) 0.009
 - b) 0.018
 - c) 0.027
 - d) 0.036
11. If one mole of a substance is present in one kg of solvent, then (E)
 - a) It shows molar concentration
 - b) It shows molal concentration
 - c) It shows normality
 - d) It shows strength g/g
12. A binary solution has two components 'A' and 'B'. The mole fraction of component 'A' is 0.5, then the number of moles of components 'A and 'B' in the solution is (A)
 - a) $n_A > n_B$
 - b) $n_A < n_B$
 - c) $n_A = n_B$
 - d) All of these
13. The concentration of fluoride ions to be present in water to prevent tooth decay is (E)
 - a) 1 ppm
 - b) 1.5 ppm
 - c) 2 ppm
 - d) 2.5 ppm
14. The concentration of pollutants in water or atmosphere is expressed in terms of (E)
 - a) $\mu\text{g L}^{-1}$
 - b) molarity
 - c) molality
 - d) volume percentage (v/v)
15. Match the following concentration term of solution with their units: (A)

	Quantity		Units
A.	molarity	p.	gmol^{-1}
B.	molality	q.	K kg/mol
C.	mole fraction	r.	Mol/kg
D.	molar mass	s.	mol/dm^3
E.	Ebullioscopic constant/ Cryscopic constant	t.	No unit

- a) A - s; B - r; C - t; D - p; E - q b) A - s; B - t; C - p; D - r; E - q
- c) A - s; B - r; C - p; D - t; E - q d) A - r; B - s; C - q; D - t; E - p

1.3 Solubility;

16. The tanks used by most of scuba divers are filled with air diluted with helium of around (E)
- a) 88.3% b) 56.2%
- c) 32.1% d) 11.7%
17. A student took two glasses of pure water from a water filter. He cools glass-A in a fridge and warms the other glass – B on a stove. On comparing the solubility of oxygen in H₂O in glass-A and glass-B, he states that glass- ‘A’ contains (A)
- a) more oxygen than glass-B. b) less oxygen than glass-B.
- c) same amount of oxygen as in glass-B. d) zero concentration of oxygen.

1.4 Vapor Pressure of liquid Solutions;

18. The dissolution of gas in a liquid is governed (E)
 - a) Raoult's law
 - b) Henry's law
 - c) Dalton's law of pressure
 - d) van't Hoff factor
19. P_A and P_B are the vapour pressures of pure liquid components, A and B, respectively of an ideal binary solution. If X_A represents the mole fraction of component A, the total pressure of the solution will be (A)
 - a) $P_A + X_A (P_B - P_A)$
 - b) $P_A + X_A (P_A - P_B)$
 - c) $P_B + X_A (P_B - P_A)$
 - d) $P_B + X_A (P_A - P_B)$
20. The vapour pressure of two liquids P and Q are 80 and 60 torr, respectively. The total vapour pressure of solution obtained by mixing 3 mole of P and 2 mol of Q would be (D)
 - a) 72 torr
 - b) 140 torr
 - c) 68 torr
 - d) 20 torr
21. Which of the following statements about the composition of the vapour over an ideal 1: 1 molar mixture of benzene and toluene is correct? Assume that the temperature is constant at 25°C. (Given, vapour pressure data at 25°C, benzene 12.8 kPa, toluene = 3.85 kPa). (A)
 - a) The vapour will contain equal amounts of benzene and toluene.
 - b) Not enough information is given to make a prediction.
 - c) The vapour will contain a higher percentage of benzene.
 - d) The vapour will contain a higher percentage of toluene.
22. An aqueous solution is 1.00 molal in KI. Which change will cause the vapour pressure of the solution to increase? (A)
 - a) Addition of NaCl
 - b) Addition of Na_2SO_4
 - c) Addition of 1.00 molal KI
 - d) Addition of water

1.5 Ideal and Non-ideal Solutions;

23. The mixture which shows positive deviation from Raoult's Law is [NEET 2020] (E)
 - a) n-hexane and n-heptane
 - b) Bromoethane and chloroethane
 - c) Ethanol and Acetone
 - d) Chloroform and Acetone
24. The mixture which shows positive deviation from Raoult's law is (E)
 - a) ethanol + acetone
 - b) benzene + toluene
 - c) acetone + chloroform
 - d) chloroethane + bromoethane.
25. The mixture that forms maximum boiling azeotrope is (E)
 - a) heptane + octane
 - b) water + nitric acid
 - c) ethanol + water
 - d) acetone + carbon disulphide.
26. Which of the following statements is correct regarding a solution of two components A and B exhibiting positive deviation from ideal behaviour? (E)
 - a) Intermolecular attractive forces between A-A and B-B are stronger than those between A-B.
 - b) $\Delta V_{\text{mix}}H = 0$ at constant T and P.
 - c) $\Delta V_{\text{mix}}V = 0$ at constant T and P.
 - d) Intermolecular attractive forces between A-A and B-B are equal to those between A-B.
27. A solution containing components A and B follows Raoult's law. (E)
 - a) A-B attraction force is greater than A - A and B-B
 - b) A-B attraction force is less than A - A and B - B
 - c) A-B attraction force remains same as A - A and B-B
 - d) volume of solution is different from sum of volume of solute and solvent.
28. All form ideal solution except (E)
 - a) C_6H_6 and $\text{C}_6\text{H}_5\text{CH}_3$
 - b) C_2H_6 and $\text{C}_2\text{H}_5\text{I}$
 - c) $\text{C}_6\text{H}_5\text{Cl}$ and $\text{C}_6\text{H}_5\text{Br}$
 - d) $\text{C}_2\text{H}_5\text{I}$ and $\text{C}_2\text{H}_5\text{OH}$
29. An ideal solution is formed when its components (E)
 - a) have no volume change on mixing.
 - b) have no enthalpy change on mixing.
 - c) have both the above characteristics.
 - d) have high solubility.
30. 10 ml of liquid 'A' and 20 ml of liquid 'B' are mixed at 25°C . The volume of the solution was measured to be 30.1 ml then, (A)
 - a) $\Delta H_{\text{mix}} > 0$, Solution shows negative deviation from Raoult's law.
 - b) $\Delta H_{\text{mix}} < 0$, Solution shows negative deviation from Raoult's law.
 - c) $\Delta H_{\text{mix}} > 0$, Solution shows positive deviation from Raoult's law.
 - d) $\Delta H_{\text{mix}} < 0$, Solution shows positive deviation from Raoult's law.
31. The mixture that forms maximum boiling azeotrope at specific concentration is (E)
 - a) heptane + octane
 - b) water + Nitric acid
 - c) ethanol + water
 - d) acetone + carbon disulphide

32. Match the following Type of liquid - liquid solution with their examples:

(A)

	Type of liquid - liquid solution		Example
A.	Ideal solution	p.	ethanol - acetone
B.	Non ideal solution with positive deviation	q.	acetone - chloroform
C.	Non ideal solution with negative deviation	r.	benzene – toluene

a) A - p; B - q; C - r;

b) A - p; B - r; C - q;

c) A - q; B - r; C - p;

d) A - r; B - p; C - q;

1.6 Colligative Properties and Determination of Molar mass;

33. In a solution containing non-volatile solute, the mole fraction of solvent is 0.9. The relative lowering of vapour pressure is

(A)

a) 1

b) 0.1

c) 0.9

d) 1.1

34. During depression of freezing point in a solution the following are in equilibrium

(A)

a) Liquid solvent, solid solvent

b) Liquid solvent, solid solute

c) Liquid solute, solid solute

d) Liquid solute, solid solvent

35. At 25°C the highest osmotic pressure is exhibited by 0.1 M solution of (assume complete dissociation)

(E)

a) Glucose

b) urea

c) CaCl_2

d) KCl

36. Of the following 0.10 m aqueous solutions, which one will exhibit the largest freezing point depression? (assume complete dissociation)

(E)

a) KCl

b) $\text{C}_6\text{H}_{12}\text{O}_6$ c) $\text{Al}_2(\text{SO}_4)_3$ d) K_2SO_4

37. Which of the following aqueous solution has minimum freezing point? (Assume complete dissociation)

(E)

a) 0.01 NaCl

b) 0.005m $\text{C}_2\text{H}_5\text{OH}$ c) 0.005m MgI_2 d) 0.005m MgSO_4

38. For 0.1 M solution, the colligative property will follow the order (assume complete dissociation)

(E)

a) $\text{NaCl} > \text{Na}_2\text{SO}_4 > \text{Na}_3\text{PO}_4$ b) $\text{NaCl} < \text{Na}_2\text{SO}_4 < \text{Na}_3\text{PO}_4$ c) $\text{NaCl} > \text{Na}_2\text{SO}_4 \approx \text{Na}_3\text{PO}_4$ d) $\text{NaCl} < \text{Na}_2\text{SO}_4 = \text{Na}_3\text{PO}_4$

39. On assuming complete dissociation, the aqueous solution having highest Freezing point is

(E)

a) 2M KCl, ($i = 2$)b) 2M K_2SO_4 , ($i = 3$)c) 2M $\text{Al}_2(\text{SO}_4)_3$, ($i = 5$)d) 2M $\text{K}_3[\text{Fe}(\text{CN})_6]$, ($i = 4$)

40. When compared to ΔT_f -value of a **0.01M** glucose solution, the ΔT_f - value of **0.01M MgCl_2** solution
a) increases about three times b) increases about six times **(A)**
c) increases about ten times d) remains same
41. Which of the following compounds can be used as antifreeze in automobile radiators? **(E)**
a) Methyl alcohol b) Nitrophenol
c) Glycol d) Ethyl alcohol
42. If molality of the dilute solution is doubled, the value of molal depression constant (K_f) will be **(A)**
a) halved b) tripled
c) unchanged d) doubled.
43. When mercuric iodide is added to the aqueous solutions of potassium iodide then: **(D)**
a) freezing point is raised. b) freezing point is lowered.
c) freezing point does not change. d) boiling point does not change
44. If molality of the dilute solution is doubled, the value of the molal depression constant (K_f) will be **(E)**
a) doubled b) halved
c) tripled d) unchanged
45. As a result of osmosis, the volume of the solution: **(E)**
a) gradually decreases b) gradually increases
c) is not affected d) suddenly increases
46. Pure water can be obtained from sea water by **(E)**
a) centrifugation b) plasmolysis
c) reverse osmosis d) sedimentation.
47. The relationship between osmotic pressure at 273 K when 10 g glucose (p_1), 10 g urea (p_2), and 10 g sucrose (p_3) are dissolved in 250 mL of water is. **(A)**
a) $P_2 > P_1 > P_3$ b) $P_2 > P_3 > P_1$
c) $P_1 > P_2 > P_3$ d) $P_3 > P_1 > P_2$
48. From the colligative properties of solution, which one is the best method for the determination of molecular weight of proteins and polymers? **(E)**
a) Osmotic pressure b) Lowering in vapour pressure
c) Lowering in freezing point d) Elevation in boiling point
49. Blood cells retain their normal shape in solution which are **(E)**
a) hypotonic to blood b) isotonic to blood
c) hypertonic to blood d) not equimolar to blood.
50. During osmosis, flow of water through a semipermeable membrane is **(E)**
a) from solution having lower concentration only
b) from solution having higher concentration only
c) from both sides of semipermeable membrane with equal flow rates
d) from both sides of semipermeable membrane with unequal flow rates.

51. Solution (A) containing FeCl_3 is separated from solution (B) containing $\text{K}_4\text{Fe}(\text{CN})_6$ by a semipermeable membrane. If FeCl_3 on reaction with $\text{K}_4[\text{Fe}(\text{CN})_6]$ produces blue colour of $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$, the blue colour will appear in (D)

- a) A b) B
c) In both A and B d) Neither in A nor in B

52. During osmosis, the solvent molecules are moving from (E)

- a) Hypotonic solution to hypertonic solution
b) Hypertonic solution to hypotonic solution
c) Higher concentrated solution to lower concentrated solution
d) Higher osmotic pressure solution to lower osmotic pressure solution.

53. Match the following **Colligative property with their formulae:** (A)

Colligative property			Formula
A.	Relative lowering of vapour pressure of the solvent	i.	$= K_f m$
B.	Henry's law	ii.	$= \frac{R \times M_1 \times T_f^2}{1000 \times \Delta_{\text{fus}} H}$
C.	Depression in freezing point	iii.	$\frac{p_1^0 - p_1}{p_1^0} = \frac{n_2}{n_1 + n_2}$
D.	Cryoscopic constant	iv.	$p = K_H \cdot x$

- a) A - iii; B - iv; C - ii; D - i b) A - ii; B - iv; C - iii; D - i
c) A - iii; B - iv; C - i; D - ii d) A - ii; B - iv; C - i; D - iii

54. Match the following **Colligative property with their formulae:** (A)

Colligative property		Formula	
A.	Raoult's law	i.	$= \frac{R \times M_1 \times T_b^2}{1000 \times \Delta_{\text{vap}} H}$
B.	Elevation of boiling point	ii.	$= CRT$
C.	Ebullioscopic constant	iii.	$p = x_1 p_1^0 + x_2 p_2^0$
D.	Osmotic pressure	iv.	$= K_b m$

- (a) A - iii; B - iv; C - ii; D - i (b) A - ii; B - iv; C - iii; D - i
(c) A - iii; B - iv; C - i; D - ii (d) A - ii; B - iv; C - i; D - iii

1.7 Abnormal Molar Masses

55. Which of the following salts has the same value of Van't Hoff factor (i) as that of $\text{K}_3[\text{Fe}(\text{CN})_6]$? (A)

- a) Na_2SO_4 b) $\text{Al}(\text{NO}_3)_3$
c) $\text{Al}_2(\text{SO}_4)_3$ d) NaCl